

DAY - 14

SEAT NUMBER

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2024 VIII 02

1100

V - 834

(E)

**COMPUTER SCIENCE
PAPER - I (D-9)**

Time : 3 Hours

4 Pages

Max. Marks : 50

Instructions : (1) All questions are compulsory.

(2) Figures to the right indicate full marks.

(3) Use of any type of calculator not allowed.

(4) Draw neat diagram wherever necessary.

1. (A) Select the correct option from the following and rewrite sentence :

(a) LINUX is _____ software. 1

(i) Free

(ii) Public Domain

(iii) Private

(iv) Paid

(b) The elements of record are _____ 1

(i) Homogenous

(ii) Similar

(iii) Non-homogenous

(iv) Identical

(c) HTML stands for _____ 1

(i) Hyper Text Markup Language

(ii) High Text Manipulation Language

(iii) Hyper Text Manipulation Language

(iv) High Text Markup Language

- (d) A pointer is a variable that holds _____ of another variable. 1
- (i) Value
 - (ii) Memory address
 - (iii) Data-type
 - (iv) None of these
- (B) Answer any two of the following :
- (a) What is GUI ? Explain any two essential components of GUI. 3
 - (b) State any six principal advantages of object oriented programming. 3
 - (c) State characteristics of friend function. 3
2. (A) Answer any two of the following :
- (a) Give the structure of HTML WebPage. 3
 - (b) Explain following tags of html with example : 3
 - (i)
 - (ii) <U>
 - (iii) <HR>
 - (c) What is Class ? Explain general form of class declaration. 3
- (B) Answer any one of the following :
- (a) Write names of two softwares for writing html code and write names of two browser software to execute html code. 4
 - (b) Explain Constructor and Destructor with example of each. 4
3. (A) Answer any two of the following :
- (a) Explain any six data structure operations. 3
 - (b) Explain with flowcharts the following control structures : 3
 - (i) Sequence logic
 - (ii) Selection logic
 - (iii) Iteration logic
 - (c) Define Security. Explain different elements of Security. 3

(B) Answer any one of the following :

(a) What is Searching ? Write an algorithm for linear search with example.

4

(b) Write any eight rules for virtual function that satisfies the compiler requirements.

4

4. (A) Answer any two of the following :

(a) Discuss virus detection, virus removal and virus prevention philosophies.

3

(b) What functions are performed by memory management system ? State any four memory management system.

3

(c) What is pointer in C++ ? Give suitable example.

3

(B) Answer any one of the following :

(a) Explain the following terms :

4

(i) Linked List

(ii) Binary Tree

(iii) STACK

(iv) QUEUE

(b) Explain following terms with respect to process management :

4

(i) Internal Priority

(ii) External Priority

(iii) Purchase Priority

(iv) Time Slice

5. Answer any two of the following :

(a) Write a program in C++ to calculate area of circle using class BSC and member function Print().

5

(b) Write a program in C++ to read the array of 10 integers. Find and print the sum of all the elements of given array.

5

(c) Write HTML code to display following output :

5

Library Books		
Section	Reference	Text Book
	2000	4000

OR

5. Attempt any two of following :

(a) Write a program in C++ to generate and print the following pattern : 5

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

(b) Write a C++ program to find the greatest common divisor of two numbers. Define a member function find() to accept the values and calculate GCD of two numbers and define print() function to print the GCD value. 5

(c) Write HTML code for following output : 5

☐ Computer Science

• Paper-I

i. C++ Language

ii. HTML

iii. OS

iv. Ds

○ Paper-II

1. Microprocessor

2. Microcontroller

3. X-86 processor

4. Networking